LARA — Autonomous Improvement Loop

Slice: **venmo-other-5** | Max fixes: **1** | Started: **2026-06-03 13:34:38**

Stage 1+2 — Baseline run + failure analysis 13:34:38

Run the pinned slice on the current code, then tag each failed task with failure-category codes.

Baseline results score=63.6

Task	Result	Assertions	Failure categories
22cc237_1	❌ WRONG	3/4 (75%)	E-PAGINATION-MISSED, E-PLAN-WRONG-API, E-WRONG-ENTITY
22cc237_2	❌ WRONG	2/4 (50%)	E-EXEC-FABRICATE, E-EXEC-INVENT-API, E-LINEAR-RETRY, E-PAGINATION-MISSED, E-PLAN-DATE-NOT-WIRED, E-PLAN-NO-CONTACT-LOOKUP, E-WRONG-ENTITY
d0b1f43_1	❌ WRONG	1/2 (50%)	
3c13f5a_1	❌ WRONG	4/6 (67%)	E-EXEC-INVENT-API, E-EXEC-WRONG-TOKEN, E-PAGINATION-MISSED
3c13f5a_2	❌ WRONG	4/6 (67%)	E-EXEC-INVENT-API, E-EXEC-WRONG-TOKEN, E-PAGINATION-MISSED, E-PLAN-WRONG-API

Stage 3 — Prioritize candidate fixes 13:41:40

Score every proposed fix: priority = impact×2 + cluster×2 + tractability×3. Pick the highest.

Ranked fix candidates priority = impact×2 + cluster×2 + tractability×3

Fix ID	Priority	Breakdown	Task hits	Surface (file)	Category
`E-WRONG-ENTITY:0`	31	I=3 C=5 T=5	2	explorer_prompt	Plan picked the wrong entity type (e.g. payment_requests for a 'payments I recei
`E-WRONG-ENTITY:1`	23	I=3 C=1 T=5	2	venmo_specialist	Plan picked the wrong entity type (e.g. payment_requests for a 'payments I recei

🟩 highlighted row = the fix the loop selected (highest priority).

💡 Why the loop chose E-WRONG-ENTITY:0

Failure category targeted: E-WRONG-ENTITY — Plan picked the wrong entity type (e.g. payment_requests for a 'payments I received' task that needs transactions).

This category hit 2 task(s) in the baseline run → impact=3.

Priority score = 31 (impact 3×2 + cluster 5×2 + tractability 5×3) — the highest of all proposed fixes.

What the fix does: Add a VENMO PATTERNS block disambiguating: 'payments I received' (past tense, settled transactions) → show_transactions(direction='received'); 'payment requests' (pending, awaiting approval) → show_received_payment_requests. List the action verbs that distinguish them.

Rationale (from fix_mappings.json): Highest priority outstanding fix (priority A in 2026-05-22 prioritization, score 35 under safety-driven weights). Solves the afc0fce family root cause.

Stage 4 — Generate patch, verify, apply 13:41:40

GPT-4.1-mini rewrites the marked region. Verify (preservation + compile), then apply git-protected.

📝 Patch proposed by GPT-4.1-mini 13:42:06

File changed: `/home/omer2/LARA_project/LARA-sub_agents/LARA/prompts.py`
Region (marker): `explorer_prompt:semantic_api_selection`
Patch cost: $0.0045 | −0 lines removed +0 lines added

Before (current code)

```
SEMANTIC API SELECTION — common task patterns:  "songs in my playlists"       → show_playlist_library → show_playlist for EACH playlist  "song librar
```

After (GPT patch)

```
SEMANTIC API SELECTION — common task patterns:  "songs in my playlists"       → show_playlist_library → show_playlist for EACH playlist  "song librar
```

✅ The patch was purely additive — no existing lines were removed.

Stage 4d + 5 — Re-run + keep-or-revert 13:42:06

Re-run the same slice with the patched code, compare scores, keep the fix only if the score improved.

Results after patch score=69.1

Task	Result	Assertions	Failure categories
22cc237_1	❌ WRONG	2/4 (50%)	E-EXEC-INVENT-API, E-EXEC-WRONG-TOKEN, E-PAGINATION-MISSED, E-PLAN-DATE-NOT-WIRED
22cc237_2	❌ WRONG	2/4 (50%)	E-PAGINATION-MISSED, E-WRONG-ENTITY
d0b1f43_1	✅ PASS	2/2 (100%)	
3c13f5a_1	❌ WRONG	2/6 (33%)	E-EXEC-WRONG-TOKEN, E-LINEAR-RETRY, E-PAGINATION-MISSED
3c13f5a_2	❌ WRONG	5/6 (83%)	E-EXEC-FABRICATE, E-EXEC-WRONG-TOKEN, E-PAGINATION-MISSED, E-PLAN-WRONG-API

✅ **KEPT — applied permanently** 13:47:27
Composite score: 63.6 → 69.1 (+5.5)
Tasks correct: 0 → 1

💡 Keep-or-revert rule

score = tasks_correct × 10 + assertions_pass_pct. The fix is kept only if the new score is **strictly greater** than the baseline. Here the score improved, so the change stays and becomes the new baseline.

Loop complete — 12.8 min, LLM cost $0.0045